



Biogeography of the grasses (Poaceae): a phylogenetic approach to reveal evolutionary history in geographical space and geological time

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The grasses (Poaceae) are the fifth most diverse family of angiosperms, including 800 genera and more than 10 000 species. Few phylogenetic studies have tried to investigate palaeo-biogeographical and palaeo-ecological scenarios that may have led to present-day distribution and diversity of grasses at the family level. We produced a dated phylogenetic tree based on combined plastid DNA sequences and a comprehensive sample of Poaceae. Furthermore, we produced an additional tree using a supermatrix of morphological and molecular data that included all 800 grass genera so that ancestral biogeography and ecological habitats could be inferred. We used a likelihood-based method, which allows the estimation of ancestral polymorphism in both biogeographical and ecological analyses for large data sets. The origin of Poaceae was retrieved as African and shade adapted. The crown node of the BEP + PACCMAD clade was dated at 57 Mya, in the early Eocene. Grasses dispersed to all continents by approximately 60 million years after their Gondwanan origin in the late Cretaceous. PACCMAD taxa adapted to open habitats as early as the late Eocene, a date consistent with recent phytolith fossil data for North America. C₄ photosynthesis first originated in Africa, at least for Chloridoideae in the Eocene at c. 30 Mya. The BEP clade members adapted to open habitats later than PACCMAD members; this was inferred to occur in Eurasia in the Oligocene. © 2010 The Linnean Society of London, *Botanical Journal of the Linnean Society*, 2010, **162**, 543–557.

ADDITIONAL KEYWORDS: ancestral reconstructions – grass family – macro-evolution – palaeo-biogeography – palaeo-ecology – phylogenetics.

INTRODUCTION

Within the angiosperms, the grasses (Poaceae) constitute the fifth most diverse family, including 800 genera and more than 10 000 species (Renvoize & Clayton, 1992; Watson & Dallwitz, 1992). Economically, Poaceae are by far the most valuable of all plant families. They are also ecologically important, dominating several of the natural and artificial landscapes of the world (Bremer, 2002). It is difficult to infer detailed patterns of grass evolution from existing

phylogenetic analyses owing to poor taxon sampling (Kellogg, 2000; GPWG, 2001). However, phylogenetic studies have identified several well-supported major lineages which form the basis of a classification of the family at subfamily and tribal levels (GPWG, 2001; Duvall *et al.*, 2007; Sánchez-Ken *et al.*, 2007; Bouchenak-Khelladi *et al.*, 2008). Bouchenak-Khelladi *et al.* (2008) sampled 84% of recognized tribes within the family for four plastid gene regions. Confirming previous results (GPWG, 2001; Sánchez-Ken *et al.*, 2007), two major clades were retrieved: one comprising exclusively C₃ grasses (Bambusoideae, Ehrhartoideae and Pooideae; BEP) and the other containing both C₄ and C₃ grasses (Panicoideae,

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Arundinoideae, Chloridoideae, Centothecoideae, Micrairoideae, Aristidoideae and Danthonioideae; PACCMAD). The evolutionary history of grasses is complex because of the high level of polymorphism within the family in terms of metabolic pathways, geographical distribution and morphological structure.

Apart from phylogenetic reconstructions, our understanding of grass evolutionary history is based on fossil data (Crepet & Feldman, 1991; Dugas & Retallack, 1993; Jacobs, Kingston & Jacobs, 1999), isotopic evidence (Cerling *et al.*, 1997; Tipple & Pagani, 2007) and phytolith data (Prasad *et al.*, 2005; Strömberg, 2005; Strömberg *et al.*, 2007). The earliest records of grass pollen date from the Palaeocene of South America and Africa, between 60 and 55 Mya (Jacobs *et al.*, 1999). These data, along with the presence of the early diverging lineages of the grasses (GPWG, 2001) in South America and Africa, suggest a Gondwanan origin for the family. The origin of the grasses has been dated at *c.* 75 Mya (Bremer, 2002) using the molecular dating method known as non-parametric rate smoothing (NPRS; Sanderson, 1997). However, major diversification of the grasses may be more recent (Kellogg, 2000). Based on carbon isotopic composition of palaeosols (Cerling *et al.*, 1997) and fossil tooth enamel (MacFadden & Cerling, 1994), *C₄* grasses, which constitute nearly half of the total species diversity of grasses, are thought to have originated in the Eocene and diversified by 15 Mya (Christin *et al.*, 2008; Vicentini *et al.*, 2008; Bouchenak-Khelladi *et al.*, 2009) but their global expansion occurred later, *c.* 7–5 Mya (Sage & Monson, 1999). To date, there have been no attempts to reconstruct the biogeographical history of the grasses at a detailed and family-wide scale, mainly because of the absence of comprehensive phylogenetic trees. The degree of ecological polymorphism within grass genera makes ancestral ecology reconstructions inaccurate and the number of equivocal nodes high. The designation of the sister group to Poaceae has been controversial. It is likely to be *Ecdeiocoleaceae*, *Joinvilleaceae* or a group containing them both (see Hodkinson *et al.*, 2007). The GPWG (2001) recognized *Joinvilleaceae* as sister to the grasses, but the comprehensive combined analysis by Bremer (2002) identified a strongly supported group relationship between Poaceae and *Ecdeiocoleaceae*.

Reconstructing the evolutionary history of a clade relies on our ability to infer the ancestral habitats and geographical distribution of lineages, but commonly used methods for such reconstructions are inappropriate because most of them were originally developed to calculate and compare tree lengths (Hardy & Linder, 2005). Most importantly, most optimization algorithms [based on either maximum par-

simony (MP) or maximum likelihood criteria (ML), respectively] allow just one state to be inferred per character per internal node (Hardy & Linder, 2005). Other methods, such as presence coding (Mickevich & Mitter, 1981), Sankoff optimizations (Sankoff & Rousseau, 1975) and polymorphism coding (as per Hardy & Linder, 2005), have been developed in order to allow ancestors of a particular lineage to exhibit polymorphic ecological characters (see review in Hardy & Linder, 2005). However, these are parsimony based and no attempts have been made so far to implement polymorphic reconstructions in a likelihood framework (but see Galley *et al.*, 2007).

In historical biogeography, the development of likelihood-based approaches has been slow compared with that of parsimony-based methods (Ree & Smith, 2008). Character model-based methods are commonly used for inferring historical biogeography without any modification (Nepokroeff *et al.*, 2003). Using a rate matrix for transition between geographical ranges, Ree & Smith (2008) developed a method for biogeographical reconstruction that allows ancestral area inference by ML. However, this method is currently unable to account for large taxon sampling and high levels of extant polymorphism when inferring biogeographical scenarios involving an ancestral or source area. Similarly, DIVA (Ronquist, 1996) only allows 180-taxon phylogenetic trees to be analysed. The utility of available reconstruction methods using ML is limited, on the one hand, by their inability to handle large numbers of taxa and, on the other, by their inability to account for intraspecific polymorphism.

As missing species introduce bias by removing the most recent speciation events, reconstruction of the ecological or biogeographical history of a clade ideally requires the production of an evolutionary framework that includes total species richness of the clade under study (Barracough & Nee, 2001). This requirement is rarely met in species-rich taxa, as they are often poorly sampled and described and technically more challenging for phylogenetics because of their size (Hodkinson *et al.*, 2007). When comprehensive taxon sampling is not achievable, alternative approaches are required. For example, Bouchenak-Khelladi *et al.* (2009) combined a large molecular data set with a suite of morphological characters to reconstruct a comprehensive phylogenetic tree of grasses using a large descriptive morphological and ecological database that is available in the form of an interactive key (Watson & Dallwitz, 1992).

The present study draws attention to these major limitations when dealing with a species-rich group such as grasses with an approach that helps overcome these problems by allowing for ancestral and extant polymorphism within an ML framework. This paper

aims to study the temporal patterns of grass biogeography and evolution by (1) producing a dated phylogenetic tree of the family, (2) inferring a comprehensive generic-level phylogenetic tree of the grasses using a combination of molecular and morphological characters and (3) inferring the evolution of geographical ranges and reconstructing ancestral ecological habitats of grass ancestors (open vs. closed canopy; habitat moisture).

MATERIAL AND METHODS

GRASS PHYLOGENY

In order to infer a complete generic-level phylogenetic tree of the grasses, we used the 298-taxon phylogenetic tree of Bouchenak-Khelladi *et al.* (2008) and additional data from the Grass Genera of the World DELTA database (Watson & Dallwitz, 1992), within which 433 morphological, anatomical and biochemical characters are coded for 800 genera. We used MP analyses on the morphological matrix with 800 genera to generate new trees, but constrained them to have the topological backbone of the majority-rule consensus tree generated with data set II of Bouchenak-Khelladi *et al.* (2008), based on Bayesian inference of three plastid DNA regions (*rbcL*, *matK* and *trnL-F*). In this way, several genera were placed on what we believe is a reliable topology of the grasses based on DNA evidence, with the remaining genera, for which no molecular data exist, placed according to unweighted and unordered morphological, anatomical and biochemical characters, as implemented in PAUP*4.0b10 (Swofford, 2002). An initial MP search used 500 replicates of tree bisection-reconnection and kept a maximum of 15 trees per replicate. The consensus tree was then used as a starting tree for five successive rounds of swapping, which yielded 10 equally most parsimonious trees.

MOLECULAR DATING AND ANCESTRAL RECONSTRUCTIONS

Molecular dating was performed on the 298-taxon phylogenetic tree of Bouchenak-Khelladi *et al.* (2008). As the molecular data were not clock-like (likelihood ratio test: $-\ln + c$ 59 470.674, $-\ln - c$ 13 964.477, d.f. = 296, $P < 0.0001$), dating was performed using a relaxed Bayesian clock, with rates being uncorrelated and sampled from a log-normal distribution (UCLN hereafter), as implemented in BEAST v1.4.6 (Drummond & Rambaut, 2007). The UCLN model was chosen because, in simulations performed by Drummond *et al.* (2006), the 95% highest posterior densities contained the true rate between 93 and 100% of the time for individual branches (see details of the method in Drummond *et al.*, (2006); [\[evolve.zoo.ox.ac.uk/beast\]\(http://evolve.zoo.ox.ac.uk/beast\)\). The substitution model used for the three different regions was determined using hierarchical likelihood ratio tests as implemented in MODELTEST 3.06 \(Posada & Crandall, 1998\). We divided our data set into three partitions and implemented a HKY + I + G model for *rbcL* and GTR + I + G models for *matK* and *trnL-F*, each with four estimated alpha categories.](http://</p>
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Four independent fossils were used for calibration. Despite the fact that the fossil record of the grasses is poor, Crepet & Feldman (1991) identified the earliest macro-fossil remains of true grasses as typical grass spikelets from the Palaeocene–Eocene boundary (55 Mya). The spikelets have two glumes and two flowers and are characteristic of the BEP and PACCMAD clades in Poaceae (Crepet & Feldman, 1991; GPWG, 2001). Accordingly, the ancestral node of the BEP + PACCMAD clade was constrained to have a minimum age of 55 My using a log-normal prior distribution (a log-normal mean of 55.0 and a log-normal standard deviation of 1.0). Additionally, two fossil grasses, from south-western Kenya identified and dated by Dugas & Retallack (1993) as members of subfamilies Panicoideae and Chloridoideae, are dated at c. 14 My (middle Miocene). The first fossil was identified as a member of Panicoideae and close to the genus *Cleistochloa*, based on the occurrence of dumbbell-shaped silica bodies with parallel low-domed subsidiary cells (Watson & Dallwitz, 1989). The second fossil was identified as a member of Chloridoideae based on the combination of round silica bodies and triangular stomatal subsidiary cells (Dugas & Retallack, 1993). The authors narrowed the identification to *Distichlis* by comparing the arrangement of the silica bodies and the distribution of papillae and stomata, with those on the epidermis of two extant *Distichlis* spp. (Dugas & Retallack, 1993). These dates were attached according to the position of these genera in the complete generic-level phylogenetic tree and then pinpointed on the 298-taxon phylogenetic tree with a minimum age of 14 My using a log-normal prior distribution (a log-normal mean of 14.0 and a log-normal standard deviation of 1.0).

Based on grass fossils found in titanosaur coprolites, Prasad *et al.* (2005) suggested that most modern grass lineages had differentiated by 65 Mya. This suggestion is questionable because it contradicts strongly earlier phylogenetic evidence (e.g. the origin of Chloridoideae is estimated to have occurred between 30 and 25 Mya; Christin *et al.*, 2008; Vicentini *et al.*, 2008; Bouchenak-Khelladi *et al.*, 2009). In order to accommodate these fossils, we supplied a minimum age for the crown group of grasses at 90 My old because the 'spikelet' grass lineages are estimated to have diversified between 20 and 25 My after the divergence of Anomochlooideae and Puelioideae

(Bremer, 2002; Christin *et al.*, 2008; Vicentini *et al.*, 2008; Bouchenak-Khelladi *et al.*, 2009). We performed two runs of 10 000 000 chains each, sampling every 1000 generations. We viewed statistics of each run using TRACER (Rambaut & Drummond, 2007) and, after removing 5 000 000 burn-in samples, we summarized the results using a majority-rule consensus tree.

GEOGRAPHICAL AND ECOLOGICAL DATA

The World Grass Genera database (Watson & Dallwitz, 1992) was used to score all 800 extant grass genera for geographical distribution and light (open or closed canopy) and soil moisture preferences. All three traits were coded as presence (1) or absence (0) for the 800 extant genera. For geographical coding, we used six areas: (1) Africa, (2) Eurasia, (3) Australia, New Zealand and Pacific Islands, (4) South-East Asia, (5) North and Central America and (6) South America. Two states were used for 'vegetation type': (1) open and (2) closed and, for 'habitat moisture', four states were used: (1) hydrophytic, (2) helophytic, (3) mesophytic and (4) xerophytic. Open habitat grasses are those that grow in environments without a canopy of other plants. In contrast, closed habitat grasses are adapted to lower light conditions and are found in habitats with at least a partial canopy. Hydrophytes are adapted to aquatic environments and helophytes to wet but marshy habitats (McDonald, Galwey & Colmer, 2002). Mesophytic plants are adapted to soils of moderate moisture content and xerophytic plants to arid and low water availability soils (McDonald *et al.*, 2002). In terms of distribution, 60 grass genera were cosmopolitan and 521 were found in only one of the coded geographical areas. For light preference (i.e. open vs. shade habitats), 346 genera had missing data and 185 were polymorphic. For edaphic moisture preference (hydrophytic, helophytic, mesophytic and xerophytic), 300 genera had missing data and 65 were polymorphic.

Under ML, ancestral states (i.e. internal node states) are estimated in such a way that the likelihood of the observed data (i.e. terminal nodes states) is maximized given the tree (topology and branch lengths) and a particular model of character evolution (Pagel, 1999). If any taxa are polymorphic for a particular character, ML methods cannot be performed because there are not any current implementations. Therefore, we developed a method, which allows for ancestral and extant polymorphism within a ML framework. It is built on that of Galley *et al.* (2007), in which each character is decomposed into a series of presence/absence characters describing each state. Each independent character was then reconstructed onto the comprehensive generic-level phylogenetic

tree with all branches equal to 1.0, using the Mk1 model of evolution (Lewis, 2001), as implemented in MESQUITE v2.6 (Maddison & Maddison, 2009), for discrete unordered characters. This can be referred to a punctuated model of evolution, in which the change of states takes place at nodes rather than along branches. In order to apply a gradual model of evolution, we also reconstructed ancestral states using the 298-taxon chronogram in which branch length information is available. In this case, the probability of character change is assumed to be proportional to branch length. Finally, we also applied a punctuated model using the 298-taxon chronogram where branches were set to 1.0.

These reconstruction methods are carried out separately for each geographical area, vegetation type and habitat moisture state, which allows the transition-rate parameter and the presence/absence probabilities at each node to be estimated independently for each character state. We did not apply a two-rate model of evolution because the one-rate model handles trees with few transitions and an imbalance of character states better than the two-rate model (Schluter *et al.*, 1997). The likelihoods for presence vs. absence of each character at 46 nodes (i.e. from the root up to the ancestral nodes of the major lineages within subfamilies) were recorded. Thereafter, the joint probability of each possible combination of states was calculated as shown in Table 1. Instead of taking into account individual likelihood values for each character, we compiled likelihood values for each character into a joint probability table (Table 1). The most likely combination(s) were then mapped onto the 298-taxon chronogram. Only optimizations with a likelihood percentage higher than 50% were used for inferring ancestral biogeography and ecology. We chose this 50% threshold to keep as much information as possible even although the achievement of statistical robustness required higher likelihood percentages.

Finally, we used Bremer's ancestral analysis (Bremer, 1992) using the same geographical distribution coding scheme as described above, to infer the ancestral area for Poaceae as a whole.

RESULTS

COMPLETE GENERIC LEVEL PHYLOGENETIC TREE

Of the 433 morphological and anatomical characters, 374 were potentially parsimony informative, 54 were uninformative and five constant. After five rounds of TBR swapping, the 10 shortest trees had 10 665 steps with a consistency index of 0.063 and a retention index of 0.605. The strict consensus tree is available online in the TREEBASE website (<http://>

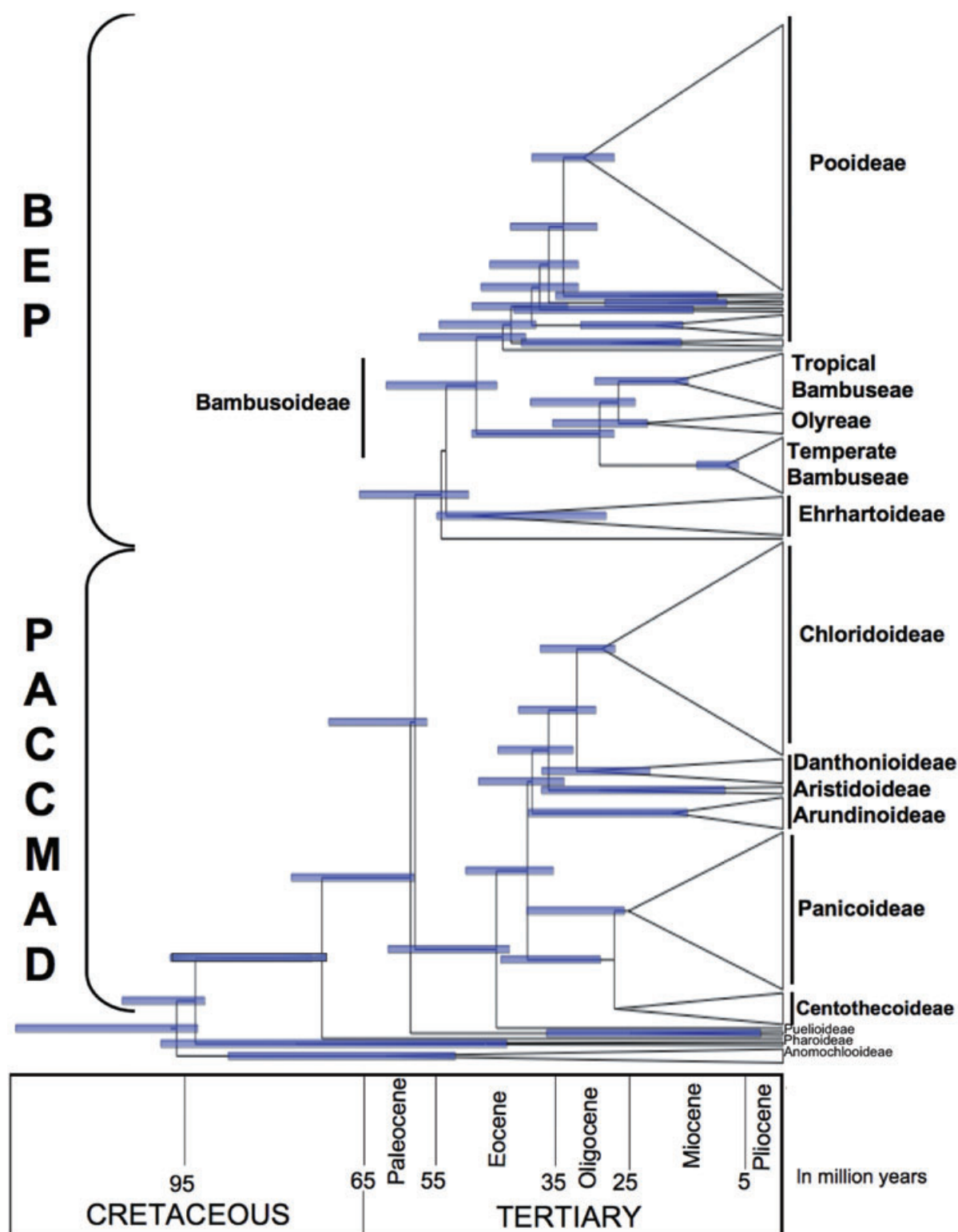


Figure 1. Two hundred and ninety-eight-grass taxa chronogram based on a relaxed Bayesian clock, with rates being uncorrelated and sampled from a lognormal distribution (UCLN as in Drummond *et al.*, 2006). BEP and PACCMAD clades following the GPWG (2001) and Sánchez-Ken *et al.* (2007), respectively. Horizontal thick bars represent 95% confidence intervals of divergence times. Note that the sizes of terminal triangles are proportional to the number of taxa imbedded in the clades.

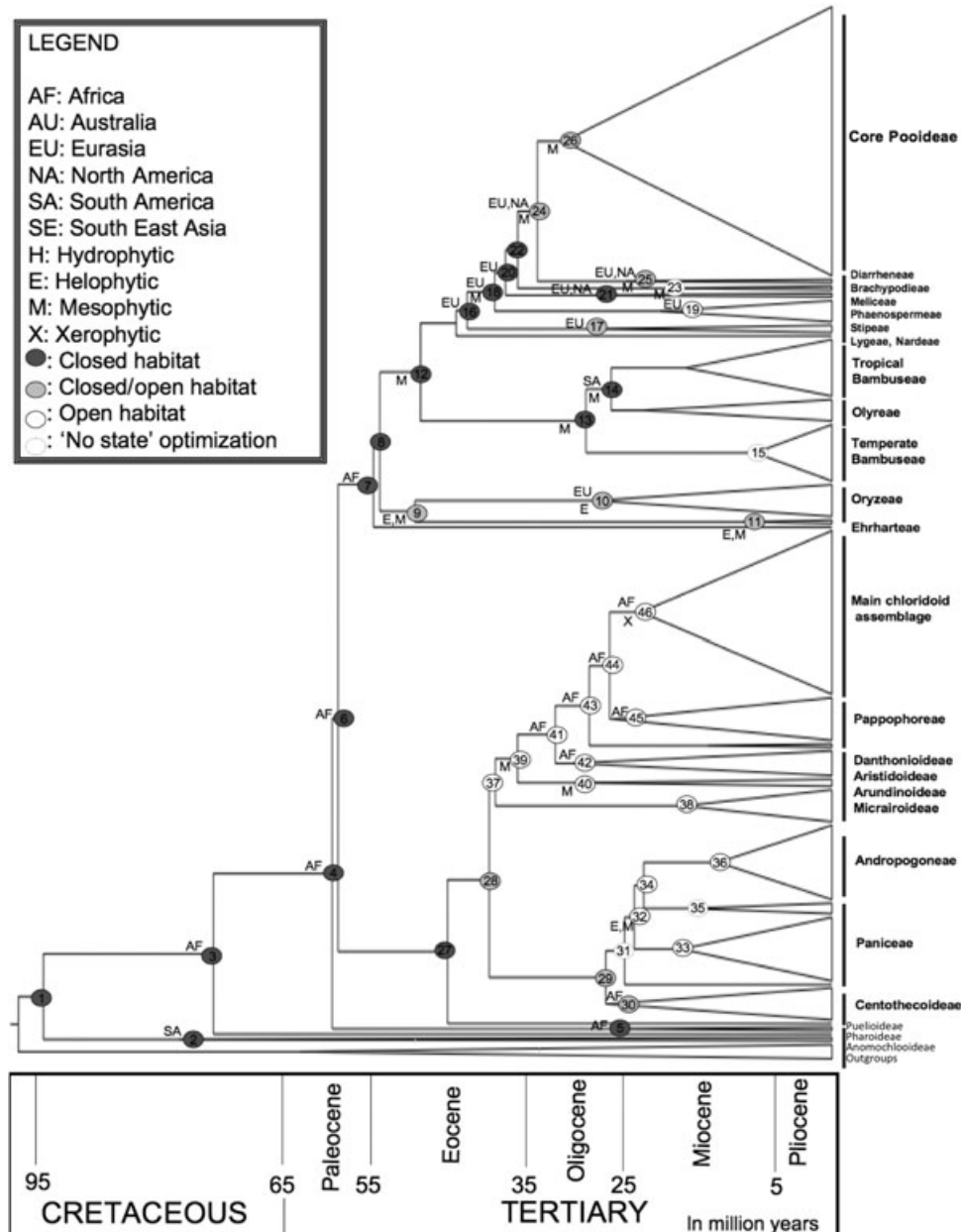


Figure 2. Ancestral reconstructions of geographical areas, vegetation type and habitat moisture using polymorphism coding and maximum likelihood (as shown in Table 1), using a comprehensive generic-level phylogenetic tree of the grasses. Likelihood estimates can be found in Table 2. Main chloridoid assemblage and core Pooideae according to Watson & Dallwitz (1992). Numbers within circles refer to node numbers in Table 2.

trees to check for consistency and no incongruences in ancestral reconstructions were found for the 46 nodes (data not shown). Because of the lack of branch length information on the comprehensive generic-level phylogenetic tree as a result of the use of morphological characters for which no model of evolution can be implemented, we assigned all branches to be equal to 1.0. We believe that the conclusions drawn here deal with evolutionary histories of major lineages, not

individual genera, and therefore do not affect the interpretations of our results.

Because of the presence of a relatively high proportion of missing data, optimizations were not possible for ten and four nodes for moisture and light preference, respectively. However, this did not affect the interpretation of our results. The origin of Poaceae was retrieved as closed canopy-adapted (node 1; 98.7% confidence, 98.7 hereafter) (Fig. 2; Table 2). It

Table 2. Maximum likelihood with polymorphism coding optimizations using the comprehensive generic level (no branch length and all taxa) and the 298-taxon chronogram (P, punctuated model where all branches equal to 1.0; G, gradual model where branches proportional to time) for 46 nodes. A, biogeographical optimizations with Africa (AF), Australia (AU), Eurasia (EU), North America (NA), South America (SA), South-East Asia (SE) and Cosmopolitan (COSMO). B, vegetation type optimizations. C, habitat moisture optimizations with hydrophytic (Hy), helophytic (E), mesophytic (M) and xerophytic (X). N/A, no significant optimizations

A

Nodes	Comprehensive phylogenetic tree		298-taxon P		298-taxon G		
	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	
1	N/A	N/A	N/A	N/A	N/A	N/A	
2	SA	84.2	N/A	N/A	N/A	N/A	
3	N/A	N/A	AF	99.1	N/A	N/A	
4	AF	75.0	N/A	N/A	N/A	N/A	
5	AF	93.6	AF	97.4	N/A	N/A	
6	AF	64.7	AU	95.7	N/A	N/A	
7	AF	62.8	N/A	N/A	N/A	N/A	
8	N/A	N/A	COSMO	94.7	N/A	N/A	
9	N/A	N/A	N/A	N/A	N/A	N/A	
10	EU	61.9	N/A	N/A	N/A	N/A	
11	N/A	N/A	EU	98.0	N/A	N/A	Ehrhartoideae
12	N/A	N/A	EU	99.3	N/A	N/A	
13	N/A	N/A	EU	97.1	N/A	N/A	
14	SA	88.7	AF and EU and NA	99.7	N/A	N/A	
15	N/A	N/A	N/A	N/A	N/A	N/A	Bambusoideae
16	EU	62.2	N/A	N/A	N/A	N/A	
17	EU	81.3	COSMO	99.4	N/A	N/A	
18	EU	77.8	N/A	N/A	N/A	N/A	
19	EU	73.2	AF and EU and NA and SA	90.8	N/A	N/A	
20	EU	79.1	EU and NA	87.1	N/A	N/A	Pooideae
21	EU and NA	61.1	N/A	N/A	N/A	N/A	
22	N/A	N/A	AF and EU and NA and SA	93.9	N/A	N/A	
23	N/A	N/A	N/A	N/A	N/A	N/A	
24	EU and NA	78.7	N/A	N/A	N/A	N/A	
25	EU and NA	74.8	COSMO	91.9	N/A	N/A	
26	N/A	N/A	EU and SA	94.3	N/A	N/A	
27	N/A	N/A	AU	95.9	N/A	N/A	
28	N/A	N/A	AF	84.9	N/A	N/A	
29	N/A	N/A	AU and SA	92.3	N/A	N/A	
30	AF	60.5	N/A	N/A	N/A	N/A	
31	N/A	N/A	AF and AU and SE and SA	89.2	N/A	N/A	Centothecoideae
32	N/A	N/A	N/A	N/A	N/A	N/A	Panicoideae
33	N/A	N/A	AF and AU	89.2	N/A	N/A	
34	N/A	N/A	NA and SA	85.8	N/A	N/A	
35	N/A	N/A	N/A	N/A	N/A	N/A	
36	N/A	N/A	AF and EU and SA	87.5	N/A	N/A	
37	N/A	N/A	SE	91.2	N/A	N/A	
38	N/A	N/A	N/A	N/A	N/A	N/A	
39	N/A	N/A	N/A	N/A	N/A	N/A	Aristidoideae
40	N/A	N/A	N/A	N/A	N/A	N/A	Arundinoideae
41	AF	66.3	N/A	N/A	N/A	N/A	Chloridoideae
42	AF	84.1	AF and EU and AU	97.9	N/A	N/A	Danthonioideae
43	AF	84.0	SA	92.9	N/A	N/A	
44	AF	85.2	N/A	N/A	N/A	N/A	
45	AF	75.0	AF	89.2	N/A	N/A	
46	AF	82.6	AF and AU and SE	93.5	N/A	N/A	

Table 2. Continued

B

Nodes	Comprehensive phylogenetic tree		298-taxon P		298-taxon G		
	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	
1	Closed	98.7	N/A	N/A	N/A	N/A	
2	Closed	99.8	N/A	N/A	N/A	N/A	
3	Closed	99.8	N/A	N/A	N/A	N/A	
4	Closed	99.9	N/A	N/A	N/A	N/A	
5	Closed	99.8	Closed	80.6	Closed	80.6	
6	Closed	99.2	N/A	N/A	N/A	N/A	
7	Closed	99.2	N/A	N/A	N/A	N/A	
8	Closed	95.8	N/A	N/A	N/A	N/A	
9	Closed or closed and open	49.9 or 49.9	N/A	N/A	N/A	N/A	
10	Closed or closed and open	49.9 or 50.0	Closed	85.1	Closed	85.2	Ehrhartoideae
11	Closed and open	95.7	Closed and open	96	Closed and open	96	
12	Closed	98.1	Closed	69.5	Closed	70.4	
13	Closed	98.8	N/A	N/A	N/A	N/A	
14	Closed	99.8	Closed	98	Closed	97.9	Bambusoideae
15	N/A	N/A	N/A	N/A	N/A	N/A	
16	Closed	97	N/A	N/A	N/A	N/A	
17	Closed and open	52.3	N/A	N/A	N/A	N/A	
18	Closed	68.3	N/A	N/A	N/A	N/A	
19	Open	87.3	N/A	N/A	N/A	N/A	
20	Closed	64.8	N/A	N/A	N/A	N/A	
21	Closed	58.8	N/A	N/A	N/A	N/A	Pooideae
22	Closed	62.2	N/A	N/A	N/A	N/A	
23	N/A	N/A	N/A	N/A	N/A	N/A	
24	Closed and open	61.9	Open	68.1	N/A	N/A	
25	Closed and open	58.6	N/A	N/A	N/A	N/A	
26	Closed and open	79.8	Open	92	Open	91.9	
27	Closed	93.7	N/A	N/A	N/A	N/A	
28	Closed and open	72.7	Open	73.6	Open	73.7	
29	Closed and open	81.5	Closed and open	70	Closed and open	70.1	
30	Closed and open	72.7	Closed and open	69.9	Closed and open	70.0	
31	N/A	N/A	N/A	N/A	N/A	N/A	Centothecoideae
32	Open	90.1	Closed and open	73.2	Closed and open	73.2	Panicoideae
33	Open	99.5	N/A	N/A	N/A	N/A	
34	Open	98	Closed and open	78.9	Closed and open	78.9	
35	N/A	N/A	N/A	N/A	N/A	N/A	
36	Open	99.7	Closed and open	95.2	Closed and open	95.2	
37	Open	90.5	N/A	N/A	Open	80.1	
38	Open	98.1	Closed and open	99.6	Open	99.6	
39	Open	97.8	Open	90.3	Open	90.3	
40	Open	85.7	N/A	N/A	N/A	N/A	Aristidoideae
41	Open	99.6	Open	97.5	Open	97.5	Arundinoideae
42	Open	99.9	Open	96.9	Open	96.9	Chloridoideae
43	Open	99.7	Open	99.3	Open	99.3	Danthonioideae
44	Open	99.7	Open	99.9	Open	99.6	
45	Open	98.7	Open	99.9	Open	99.9	
46	Open	99.8	Open	99.9	Open	99.9	

Table 2. *Continued*

C

Nodes	Comprehensive phylogenetic tree		298-taxon P		298-taxon G		
	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	Ancestral state	Likelihood (%)	
1	N/A	N/A	N/A	N/A	N/A	N/A	
2	N/A	N/A	N/A	N/A	N/A	N/A	
3	N/A	N/A	N/A	N/A	N/A	N/A	
4	N/A	N/A	X	99.7	N/A	N/A	
5	N/A	N/A	M	99.6	N/A	N/A	
6	N/A	N/A	M and X	99.5	N/A	N/A	
7	N/A	N/A	N/A	N/A	N/A	N/A	
8	N/A	N/A	M and X	99.5	N/A	N/A	
9	E/M	52.2	N/A	N/A	N/A	N/A	Ehrhartoideae
10	E	94.0	N/A	N/A	E/M	73.2	
11	E/M	86.6	X	99.6	N/A	N/A	
12	M	53.2	X	96.7	N/A	N/A	
13	M	70.6	M and X	99.6	N/A	N/A	Bambusoideae
14	M	84.0	X	99.7	N/A	N/A	
15	N/A	N/A	N/A	N/A	N/A	N/A	
16	N/A	N/A	N/A	N/A	N/A	N/A	Pooideae
17	N/A	N/A	M	99.7	N/A	N/A	
18	M	55.8	N/A	N/A	N/A	N/A	
19	N/A	N/A	N/A	N/A	N/A	N/A	
20	N/A	N/A	M and X	98.9	N/A	N/A	
21	N/A	N/A	N/A	N/A	N/A	N/A	
22	N/A	N/A	M and X	99.0	N/A	N/A	
23	M	63.7	N/A	N/A	N/A	N/A	
24	M	59.9	N/A	N/A	N/A	N/A	
25	M	67.1	M and X	99.3	N/A	N/A	
26	M	69.0	X	92.4	N/A	N/A	
27	N/A	N/A	M and X	99.3	N/A	N/A	Centothecoideae Panicoideae
28	N/A	N/A	M and X	99.6	N/A	N/A	
29	N/A	N/A	X	99.7	N/A	N/A	
30	N/A	N/A	N/A	N/A	N/A	N/A	
31	N/A	N/A	M and X	99.6	N/A	N/A	
32	E/M	50.8	N/A	N/A	N/A	N/A	
33	N/A	N/A	E and M	99.6	N/A	N/A	
34	N/A	N/A	N/A	N/A	N/A	N/A	
35	N/A	N/A	N/A	N/A	N/A	N/A	
36	N/A	N/A	M and X	99.0	E/M	75.1	
37	N/A	N/A	X	99.7	N/A	N/A	Aristidoideae Arundinoideae Chloridoideae Danthonioideae
38	N/A	N/A	N/A	N/A	N/A	N/A	
39	M	52.3	N/A	N/A	N/A	N/A	
40	M	70.8	N/A	N/A	N/A	N/A	
41	N/A	N/A	N/A	N/A	N/A	N/A	
42	N/A	N/A	E	99.7	N/A	N/A	
43	N/A	N/A	N/A	N/A	N/A	N/A	
44	N/A	N/A	N/A	N/A	N/A	N/A	
45	N/A	N/A	N/A	N/A	N/A	N/A	
46	X	55.3	E and M and X	99.6	N/A	N/A	

Table 3. Results of the Bremer's ancestral analysis for the six geographical areas coded. The likelihood ratio is calculated as gains/losses, the higher the ratio is, the more likely the area belongs to the ancestral area of origin (Bremer, 1992). Africa has the highest likelihood ratio and may therefore be the most likely centre of origin of the grasses

Geographical areas	Gains	Losses	Ratio
Africa	248	265	0.936
Eurasia	235	275	0.854
Australia	159	311	0.511
South-East Asia	157	330	0.476
North America	163	305	0.534
South America	194	351	0.553

is inferred to be African in Bremer's ancestral analysis (Table 3) and South American (29.6) or African (16.0) in ML optimizations (Table 2). The ancestors of Anomochlooideae were distributed in South American forests (node 2; 84.2, 99.8) (Fig. 2; Table 2). The divergences of Pharoideae and Puelioideae were optimized as being adapted to closed habitats (node 3; 99.8) (Fig. 2; Table 2). Whether these ancestors inhabited wet (i.e. hydrophytic) (32.1) or moist (i.e. helophytic) (32.9) environments is not clear based on our results (Table 2). The origin of the BEP + PACCMAD clade was in a closed environment (node 6; 99.2) (Fig. 2; Table 2).

The BEP clade still originated in closed habitats (node 7; 99.2) (Fig. 2; Table 2). The ancestors of Pooideae and Bambusoideae were reconstructed as being in closed habitats (node 12; 98.1) (Fig. 2; Table 2). The ancestral habitat of Bambusoideae was reconstructed as occurring under closed conditions (node 13; 98.8) with mesophytic preference (70.6) (Fig. 2; Table 2). Within Pooideae, tribes Lygeae + Nardeae were found to have originated in Eurasia (node 17; 81.3) (Fig. 2; Table 2). Core Pooideae were adapted to a mixed closed/open habitat (node 26; 79.8), but the centre of origin was not found to be significant even although it seems that Eurasia and North America form a likely ancestral area (Fig. 2; Table 2).

The PACCMAD clade is inferred to have originated in a closed canopy environment (node 27; 93.7) and successively to a mixed closed/open habitat (node 28; 72.7) (Fig. 2; Table 2). The ancestors of Panicoideae and Centothecoideae were similar in their habitat to the PACCMAD ancestors (node 29; 81.5) (Fig. 2; Table 2). Centothecoideae originated in a closed/open environment (node 30; 72.7) (Fig. 2; Table 2). The Aristidoideae + Chloridoideae + Danthonioideae + Micrairoideae clade originated in an open environment (node 37; 90.5) (Fig. 2; Table 2), but for Pani-

coideae the centre of origin remained uncertain (Table 2). Aristidoideae originated in an open (node 40; 85.7) and mesophytic habitat (70.8) and Danthonioideae was likely to have originated in an open African environment (node 42; 99.9, 84.1) (Fig. 2; Table 2). Finally, in Africa (node 44; 85.2), Chloridoideae diverged and adapted to an open habitat (99.7) (Fig. 2; Table 2).

Using the 298-taxon chronogram with a gradual model, we found no significant optimizations for biogeographical area (Table 2). Twenty-one and 22 out of 46 nodes were found significant for light requirement and habitat moisture, respectively (Table 2). Using a punctuated model (all branches equal to 1.0), we found 26, 21 and 22 significant optimizations for biogeography, light requirement and habitat moisture (Table 2). However, there were significant discrepancies with the optimizations carried out on the comprehensive phylogenetic tree (Table 2). Only one ancestral geographical area, 14 vegetation types (open vs. closed) and no habitat moisture were commonly retrieved with both phylogenetic trees.

DISCUSSION

RECONSTRUCTION OF POLYMORPHISM IN ANCESTRAL BIOGEOGRAPHY AND ECOLOGY

One of the major limitations with ecological reconstructions lies with the assumption of taxon monomorphism (i.e. optimization algorithms allow just one state per character per internal node), a concept that is often violated because of the amount of ecological variability within species (Hardy, 2006). To deal with polymorphism, presence coding is often used even although it may result in an artefactual reconstruction of 'no state' at some internal nodes (Hardy & Linder, 2005; Table 2). Polymorphism coding assigns the desired transition costs between the states and allows for both extant and ancestral polymorphisms for unordered discrete traits. To date, a comparable approach has not been implemented using an ML criterion. The alternative, taken in this study, was to use presence coding for ML ancestral reconstructions, compute the likelihood of all possible combinations manually (Table 1) and select the most likely combination for each internal node.

Recently, a maximum likelihood inference approach of geographical range evolution has been developed that specifies transition rates between discrete geographical areas along branches of a tree and estimates likelihoods of ancestral areas at cladogenesis events (Ree & Smith, 2008). Even although this method allows for a more robust reconstruction, within which priors regarding dispersal constraints (for instance, not allowing dispersal between Africa

and South America after 60 Mya) can be enforced, this software still cannot cope with a high degree of polymorphism in geographical distribution of extant taxa. Therefore, to apply such a method, we would have to infer species-level phylogenies of grasses because levels of polymorphism would be lower than at generic levels. However, this would not allow for the comprehensive sampling that is required for accurate ancestral reconstructions (Salisbury & Kim, 2001). Because there are no significantly optimized ancestral states, ML assigns illogical assignment of 'no state' to these nodes. It is worth noting the amount of non-significant optimizations (i.e. likelihood < 50%), especially for geographical areas and habitat moisture (22 and 20 out of 46 nodes, respectively; Table 2). This lack of confidence in ML reconstructions may be attributable to the number of character states considered (six geographical states and four habitat moisture states) and the large number of polymorphic taxa.

Using the 298-taxon chronogram, 46, 44 and 25 nodes were not optimized significantly with ML for biogeography, habitat moisture and light requirement, respectively (Table 2). The large number of missing taxa (298 instead of all 800 recognized genera) has led to non-significant optimization using ML. When using a punctuated model of evolution, most of the basal nodes were not significantly optimized and the reconstructions differ substantially when using the comprehensive generic-level phylogenetic tree (Table 2). Incomplete taxon sampling can distort branch lengths and thus affect the accuracy of ML ancestral reconstruction under a gradual model, but the effect can be even larger in a punctuated model in which a transition is optimized that may have been lost as a result of extinction or missing species.

We were not able to use DIVA (Ronquist, 1996) because the software cannot deal with sample sizes larger than 180 taxa. Bremer's ancestral analysis (BAA; Bremer, 1992) results are shown in Table 3. The ancestral area of grasses is retrieved as Africa (Table 3). Bremer's method (1992) is based on comparing the number of gains and losses of an area on the cladogram. Every event of colonization is a gain. It is assumed that the more gains (or colonization events) that are required for an area, the less likely it has been colonized and thus the more likely it was part of the ancestral area. BAA reconstructions infer Africa as the centre of origin for grasses, whereas our ML method infers South America preceding an African origin for the BEP and PACCMAD clades (Table 2). Both approaches suggest a Gondwanan origin for grasses, which can be explained by a vicariance event during the break-up of Gondwana c. 100 Mya.

PALAEO-BIOGEOGRAPHY AND PALAEO-ECOLOGY OF GRASS ANCESTORS

According to our molecular dating (Fig. 1), the ancestral lineage of what we today recognize as grasses originated in the late Cretaceous (c. 96 Mya). It is believed that terrestrial plants passed through the Cretaceous/Tertiary (K/T) boundary, with taxonomic richness being poor relative to that of today (Macleod *et al.*, 1997). Our results indicate that Poaceae originated either in the African or South American regions of Gondwana (Fig. 2; Tables 2 and 3). A Gondwanan origin is corroborated by Bremer's (2002) reconstruction of the biogeography of Poales. Subfamilies Anomochlooideae and Pharoideae have a South American and pantropical distribution, respectively. Anomochlooideae, Pharoideae and Puelioideae currently inhabit shaded tropical or warm temperate forest understories (GPWG, 2001) and their ancestral habitat was inferred as being closed canopy (Fig. 2; Table 2). The evidence therefore suggests that proto-grasses might have inhabited similar moist and forested environments. The separation of the African and South American continents occurred by the end of the Albian (96 Mya), but it has been reported that plant dispersals were common across the newly formed Atlantic until the early Tertiary through a series of islands (Morley, 2000). Grasses with spikelets include Pharoideae, Puelioideae and the BEP and PACCMAD clades (GPWG, 2001). Therefore, the appearance of one of the most striking characteristics of grasses today, grass spikelets, occurred well after their late Cretaceous origin, but still early in the evolution of the grasses at c. 75 Mya according to our reconstructions, probably in forest understories; a finding consistent with those of Kellogg (2000).

According to our molecular dating, the crown node of the BEP clade is c. 10 My older than the PACCMAD clade (53 vs. 45 My). Ancestors of the BEP + PACCMAD clade were also adapted to a forested environment (Fig. 2; Table 2). Ancestors of the BEP clade were inferred to have originated in the early Eocene in closed habitats of Africa (Fig. 2; Table 2). It is therefore plausible that ancestors of the BEP clade were part of dry woody vegetation, the group subsequently dispersing to Eurasia, starting in the middle Eocene (Fig. 2; Table 2). Pooideae are also inferred to have dispersed to Eurasia from the middle Eocene to the early Oligocene (between 47 and 38 Mya) (Fig. 2). Colonization of Eurasia by the grasses might therefore have occurred via India, which was connected to Asia by the present-day south-eastern Asian land mass, as early as the late Cretaceous–early Tertiary (Klootwijk *et al.*, 1992). Phytolith evidence from Turkey suggests the presence and relative abundance of closed-habitat grasses by

the Eocene (Strömberg *et al.*, 2007). For Pooideae, adaptation to open-habitat environments occurred long after the emergence of their major tribes, as all tribe ancestors were inferred as either closed-habitat or open/closed-habitat grasses (Fig. 2). The first dispersal within Pooideae from Eurasia was inferred to have occurred at approximately 38 Mya in the Oligocene into a mesophytic and mixed open/closed habitat of North America (Fig. 2). This scenario of a spread from the Old to the New World via the North Atlantic Land Bridge (NALB) has already been described for other tropical angiosperm clades (Chanderbali, Van Der Werff & Renner, 2001; Donoghue, 2008). Strömberg (2005) found the earliest evidence for bamboo-like grasses in the fossil record around the early Oligocene in North America, which is concordant with dispersal via the NALB. We may speculate that this migration of closed-habitat types across a latitudinal gradient might be explained by the global warming trend during the first half of the Tertiary that lead to the expansion of tropical and paratropical forests into higher latitudes (Donoghue, 2008). The crown node of Bambusoideae originated 30 Mya in the middle Oligocene (Fig. 1). Tribe Bambuseae are not monophyletic; one lineage contains tropical woody bamboos and herbaceous bamboos (Olyreae), the other includes temperate woody bamboo taxa (Bouchenak-Khelladi *et al.*, 2008; Sungkaew *et al.*, 2008). Notably, the tropical lineages of Bambuseae and Olyreae originated at *c.* 25 Mya, whereas the temperate Bambuseae were *c.* 20 My younger (Fig. 1). This divergence event seems to be correlated with two different phases of dispersal, one in the Americas and the other to and in South-East Asia, but this South-East Asian origin was not found to be significant (Fig. 2; Table 2). Both lineages have adapted to forested environments and mesophytic habitat. The spread to South America, as suggested by our geographical mapping, could have occurred along a dispersal route via the NALB until the late Eocene–early Oligocene (Tiffney & Manchester, 2001), and finally to South America, which was connected by the Greater Antilles, the Bahamas platform, the Aves Ridge (which is submerged today) and perhaps the site of the present Lesser Antilles (Rage, 1996). However, a common finding in phylogenetic studies (i.e. South-East Asian and East Asian taxa being sister to derived New World taxa, as is the case within the BEP clade in our study) often leads to the conclusion that the Bering Land Bridge (BLB) was a major dispersal route to the New World (Li *et al.*, 2000). Both NALB and BLB dispersal routes are consistent with our reconstructions.

The origin of the PACCMAD clade is estimated at *c.* 45 Mya in the middle Eocene (Fig. 1). The ancestors of this clade are inferred to be adapted to forested

habitats (Fig. 2; Table 2). Occupation of open habitats in Africa occurred in the late Eocene–early Oligocene among the ancestors of Chloridoideae and Danthonioideae (Fig. 2; Table 2). The Oligocene period was considerably drier than the rest of the Tertiary and these factors might have had an effect on the decrease of the forest cover and the expansion of open habitats (Janis, 1993). This date for the earliest major shift of grasses from closed to open habitats types is consistent with the date suggested by Strömberg (2005) based on phytolith data, adding considerable weight to the hypothesis that the late Eocene marked the early taxonomic diversification of open habitat grasses in Africa. Strömberg (2005) found that, despite this earlier taxonomic diversification of open-habitat types, typically PACCMAD grasses, a greater spread at the expense of closed-habitat types occurred during the late Oligocene and the early Miocene. It is plausible that the occupation of open habitats was possibly triggered in Africa well before they became ecologically dominant in other parts of the world, as in North America (Strömberg, 2005) or southern Europe (Strömberg *et al.*, 2007). The adaptive transition from a wetter and shaded environment to a drier open habitat is more pronounced for Chloridoideae, between 35 and 25 Mya, likely in Africa (Figs 1 and 2; Table 2). Indeed, the environmental history of Africa since the Eocene involves the expansion of arid-adapted vegetation (Bobe, 2006). Aristidoideae and Chloridoideae include exclusively C_4 grasses (except one genus and one species, respectively) and the dates of appearance are in agreement with recent studies (Christin *et al.*, 2008; Vicentini *et al.*, 2008; Bouchenak-Khelladi *et al.*, 2009), although the molecular dating techniques differed. Based on the evidence presented here, C_4 photosynthesis appears to have first originated in Africa, at least for Chloridoideae, with C_4 ancestors then dispersing to other continents. Panicoideae are inferred to have originated approximately 26 Mya in the early Miocene in mixed open/closed habitat (Figs 1 and 2). Although there is considerable uncertainty in ancestral reconstructions for this clade, it seems that Paniceae and Andropogoneae dispersed from Africa to Australia and/or North America from the early Miocene (Figs 1 and 2; Table 2). Paniceae originated relatively recently (i.e. in the middle Miocene, *c.* 15 Mya), occupying open environments (Fig. 2; Table 2). There is increasing evidence of multiple dispersals into Australia throughout the Cenozoic in many unrelated plant clades and from many directions (McLoughlin, 2001; Hopper & Gioia, 2004), but the earliest record of grasses from Australia is in a middle Eocene pollen flora (Frakes & Vickers-Rich, 1991). Australia appears to have had a dominant tropical climate during the Palaeogene followed by cooling reflecting

general global trends starting in the Oligocene (Singh, 1988). It may explain the early record of grasses adapted to tropical forest by the Eocene, followed by extinctions because of the global cooling starting in the Oligocene.

Our approach, incorporating different data sets (molecular, morphological, ecological and geographical), which have in common an overlap of taxa, has yielded a thorough analysis of grass evolution and ancestral biogeographical ranges and ecological niches. The results are enriched by palaeontological and taxonomic data. It has helped reveal macro-evolutionary patterns and has proven to be a powerful approach for dealing with species-rich group of organisms such as grasses.

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REFERENCES

- Barracough TG, Nee S. 2001.** Phylogenetics and speciation. *Trends in Ecology and Evolution* **16**: 391–399.
- Bohe R. 2006.** The evolution of arid ecosystems in eastern Africa. *Journal of Arid Environment* **66**: 564–584.
- Bouchenak-Khelladi Y, Salamin N, Savolainen V, Forest F, van der Bank M, Chase MW, Hodkinson TR. 2008.** Large multi-gene phylogenetic trees of the grasses (Poaceae): progress towards complete tribal and generic level sampling. *Molecular Phylogenetics and Evolution* **47**: 488–505.
- Bouchenak-Khelladi Y, Verboom GA, Hodkinson TR, Salamin N, Francois O, Chonghaile GN, Savolainen V. 2009.** The origins and diversification of C_4 grasses and savanna-adapted ungulates. *Global Change Biology* **15**: 2397–2417.
- Bremer K. 1992.** Ancestral areas: a cladistic reinterpretation of the center of origin concept. *Systematic Biology* **41**: 436–445.
- Bremer K. 2002.** Gondwanan evolution of the grass alliance of families (Poales). *Evolution* **56**: 1374–1387.
- Cerling TE, Harris JM, Macfadden BJ, Leakey MG, Quade J, Eisenmann V, Ehleringer JR. 1997.** Global vegetation change through the Miocene/Pliocene boundary. *Nature* **389**: 153–158.
- Chanderbali AS, Van Der Werff H, Renner SS. 2001.** Phylogeny and historical biogeography of Lauraceae: evidence from the chloroplast and nuclear genomes. *Annals of the Missouri Botanical Garden* **8**: 104–134.
- Christin PA, Besnard G, Samaritani E, Duvall MR, Hodkinson TR, Savolainen V, Salamin N. 2008.** Oligocene CO_2 decline promoted C_4 photosynthesis in grasses. *Current Biology* **18**: 1–7.
- Crepet WL, Feldman GD. 1991.** The earliest remains of grasses in the fossil record. *American Journal of Botany* **78**: 1010–1014.
- Donoghue MJ. 2008.** A phylogenetic perspective on the distribution of plant diversity. *Proceedings of the National Academy of Sciences USA* **105**: 11549–11555.
- Drummond AJ, Ho SYW, Phillips MJ, Rambaut A. 2006.** Relaxed phylogenetics and dating with confidence. *PLoS Biology* **4**: e88.
- Drummond AJ, Rambaut A. 2007.** BEAST: Bayesian Evolutionary Analysis by Sampling Trees. *BMC Evolutionary Biology* **7**: 214.
- Dugas DP, Retallack GJ. 1993.** Middle Miocene fossil grasses from Fort Ternan, Kenya. *Journal of Paleontology* **67**: 113–128.
- Duvall MR, Davis JI, Clark LG, Noll JD, Goldman DH, Sánchez-Ken G. 2007.** Phylogeny of the grasses (Poaceae) revisited. *Aliso* **23**: 237–247.
- Frakes LA, Vickers-Rich P. 1991.** Paleoclimatic setting and paleogeographic links of Australia in the Phanerozoic. In: Vickers-Rich P, Baird RF, Rich TH, eds. *Vertebrate paleontology of Australasia*. Melbourne: Pioneer Design Studios, 111–146.
- Galley C, Bytebier B, Bellstedt DU, Linder HP. 2007.** The Cape element in Afrotropical flora: from Cape to Cairo? *Proceedings of the Royal Society B* **27**: 535–543.
- GPWG. 2001.** Phylogeny and subfamilial classification of the grasses (Poaceae). *Annals of the Missouri Botanical Garden* **88**: 373–457.
- Hardy CR. 2006.** Reconstructing ancestral ecologies: challenges and possible solutions. *Diversity and Distributions* **12**: 7–19.
- Hardy CR, Linder HP. 2005.** Intraspecific variability and timing in ancestral ecology reconstructions: a test case from the Cape flora. *Systematic Biology* **54**: 299–316.
- Hodkinson TR, Savolainen V, Jacobs SW, Bouchenak-Khelladi Y, Kinney MS, Salamin N. 2007.** Supersizing: progress in documenting and understanding grass richness. In: Hodkinson TR, Parnell JAN, eds. *Reconstructing the Tree of Life: Taxonomy and systematics of species rich taxa*. Boca Raton: Taylor and Francis, CRC Press, 279–298.
- Hopper SD, Gioia P. 2004.** The Southwest Australian Floristic Region: evolution and conservation of a global hotspot of biodiversity. *Annual Review of Ecology and Systematics* **35**: 623–650.
- Jacobs BF, Kingston JD, Jacobs LL. 1999.** The origin of grass-dominated ecosystems. *Annals of the Missouri Botanical Garden* **86**: 590–643.
- Janis CM. 1993.** Tertiary mammal evolution in the context of changing climates, vegetation, and tectonic events. *Annual Review of Ecology and Systematics* **24**: 467–500.
- Kellogg EA. 2000.** The grasses: a case study in macroevolution. *Annual Review of Ecology and Systematics* **31**: 217–238.
- Klootwijk CT, Gee JS, Peirce JW, Smith GM, Macfadden PL. 1992.** An early India-Asia contact: paleomagnetic constraints from Ninetyeast Ridge, ODP Leg 121. *Geology* **20**: 395–398.

- Lewis PO. 2001.** A likelihood approach to estimating phylogeny from discrete morphological character. *Systematic Biology* **50**: 913–925.
- Li J, Bogle AL, Klein AS, Donoghue MJ. 2000.** Phylogeny and biogeography of *Hamamelis* (Hamamelidaceae). *Harvard Papers in Botany* **5**: 171–178.
- Macfadden BJ, Cerling TE. 1994.** Fossil horses, carbon isotopes and global change. *Trends in Ecology and Evolution* **9**: 481–486.
- Macleod N, Rawson PF, Forey PL, Banner FT, Boudagherfedel MK, Bown PR, Burnett JA, Chambers P, Culver S, Evans SE, Jeffery C, Kaminski MA, Lord AR, Milner AC, Morris N, Owen E, Rosen BR, Smith AB, Urguhart E, Young JR. 1997.** The Cretaceous–Tertiary biotic transition. *Journal of the Geological Society* **154**: 265–292.
- Maddison WP, Maddison DP. 2009.** Mesquite: a modular system for evolutionary analysis. Version 2.6., <http://mesquiteproject.org>.
- McDonald MP, Galwey NW, Colmer TD. 2002.** Similarity and diversity in adventitious root anatomy as related to root aeration among a range of wetland and dryland grass species. *Plant, Cell & Environment* **25**: 441–451.
- McLoughlin S. 2001.** The breakup history of Gondwana and its impact on pre-Cenozoic floristic provincialism. *Australian Journal of Botany* **49**: 271–300.
- Mickevich MF, Mitter C. 1981.** Treating polymorphic characters in systematics: A phylogenetic treatment of electrophoretic data. In: Funk VA, Brooks DR, eds. *Advances in cladistics: proceedings of the first meeting of the Willi Hennig Society*. New York: New York Botanical Garden Press, 45–58.
- Morley RJ. 2000.** *Origin and evolution of tropical rainforest*. Chichester: John Wiley & Sons, Ltd.
- Nepokroeff M, Sytsma KJ, Wagner WL, Zimmer EA. 2003.** Reconstructing ancestral patterns of colonization and dispersal in the Hawaiian understory tree genus *Psychotria* (Rubiaceae): a comparison of parsimony and likelihood approaches. *Systematic Biology* **52**: 820–838.
- Pagel M. 1999.** Inferring the historical patterns of biological evolution. *Nature* **401**: 877–884.
- Posada D, Crandall K. 1998.** Modeltest: testing the model of DNA substitution. *Bioinformatics* **14**: 817–818.
- Prasad V, Strömberg CAE, Alimohammadian H, Sahni A. 2005.** Dinosaur coprolites and the early evolution of grasses and grazers. *Science* **310**: 1177–1180.
- Rage JC. 1996.** *Terrestrial trans-Tethyan dispersals: an overview*. In First European Workshop on Vertebrate Paleontology. DGF Online Series, Vol. 1, Copenhagen: Geological Society of Denmark. http://2dgm.dk/publikationer/dgf_on_line/vol_1/rage.html.
- Rambaut A, Drummond AJ. 2007.** Tracer v1.4. Available from <http://beast.bio.ed.ac.uk/Tracer>.
- Ree RH, Smith SA. 2008.** Maximum likelihood inference of geographic range evolution by dispersal, local extinction, and cladogenesis. *Systematic Biology* **57**: 4–14.
- Renvoize SA, Clayton WD. 1992.** Classification and evolution of the grasses. In: Chapman R, ed. *Grass evolution and domestication*. Cambridge: Cambridge University Press.
- Ronquist F. 1996.** DIVA version 1.1. Computer program and manual available by anonymous FTP from Uppsala University (<ftp.uu.se> or <ftp.systbot.uu.se>).
- Sage RF, Monson RK. 1999.** *C₄ plant biology*. San Diego: Academic Press.
- Salisbury BA, Kim J. 2001.** Ancestral state estimation and taxon sampling density. *Systematic Biology* **50**: 557–564.
- Sánchez-Ken JG, Clark LG. 2007.** Phylogenetic relationships within the Centothecoidae + Panicoideae clade (Poaceae) based on *ndhF* and *rpl16* intron sequences and structural data. *Aliso* **23**: 487–502.
- Sánchez-Ken JG, Clark LG, Kellogg EA, Kay EE. 2007.** Reinstatement and emendation of subfamily Micrairoideae (Poaceae). *Systematic Botany* **32**: 71–80.
- Sanderson MJ. 1997.** A nonparametric approach to estimating divergence times in the absence of rate constancy. *Molecular Biology and Evolution* **14**: 1218–1231.
- Sankoff D, Rousseau P. 1975.** Locating the vertices of a Steiner tree in arbitrary space. *Mathematical Programming* **9**: 240–246.
- Schluter D, Price T, Mooers AO, Ludwig D. 1997.** Likelihood of ancestor states in adaptive radiation. *Evolution* **51**: 1699–1711.
- Singh A. 1988.** History of aridland vegetation and climate: a global perspective. *Biological Reviews* **63**: 156–198.
- Strömberg CAE. 2005.** Decoupled taxonomic radiation and ecological expansion of open-habitat grasses in the Cenozoic of North America. *Proceedings of the National Academy of Science USA* **102**: 11980–11984.
- Strömberg CAE, Werdelin L, Friis EM, Sarac G. 2007.** The spread of grass-dominated habitats in Turkey and surrounding areas during the Cenozoic: phytolith evidence. *Paleogeography, Paleoclimatology, Paleocology* **250**: 18–49.
- Sungkaew S, Stapleton CMA, Salamin N, Hodkinson TR. 2008.** Non-monophyly of the woody bamboos (Bambuseae; Poaceae): a multi-gene region phylogenetic analysis of Bambusoideae s.s. *Journal of Plant Research* **122**: 95–108.
- Swofford DL. 2002.** *Phylogenetic Analysis Using Parsimony (*and Other Methods)*. Version 4. Sunderland: Sinauer Associates.
- Tiffney BH, Manchester SR. 2001.** The use of geological and paleontological evidence in evaluating plant phylogeographic hypotheses in the Northern Hemisphere Tertiary. *International Journal of Plant Science* **162**: S3–S17.
- Tipple BJ, Pagani M. 2007.** The early origins of terrestrial C₄ photosynthesis. *Annual Review of Earth and Planetary Sciences* **35**: 435–461.
- Vicentini A, Barber JC, Aliscioni SS, Guissani LM, Kellogg EA. 2008.** The age of the grasses and clusters of origins of C₄ photosynthesis. *Global Change Biology* **14**: 1–15.
- Watson L, Dallwitz MJ. 1989.** Grass genera of the world: interactive identification and information retrieval. Version 1.0. *Flora Online* 22. <http://www.motbot.org/plantscience/ResBot/FO/FloraOnline.htm>.
- Watson L, Dallwitz MJ. 1992.** *The Grass Genera of the World*. Wallingford: CAB International.